DE-RENDER: QUANTUM SOULS

GRAND THEFT AUTO V • MOTH QUANTUM BOUNTY ENTRY • TECHNICAL WHITEPAPER & PIPELINE (FINAL RELEASE V1)

> *"What do we actually know about a stranger walking down the street?*
> *In Grand Theft Auto V, every pedestrian possesses hidden internal code—shooting accuracy, relationship factions, hidden cash, audio sound banks, and alertness states.*
> *In exchange for revealing their digital data, we erase their physical face."*

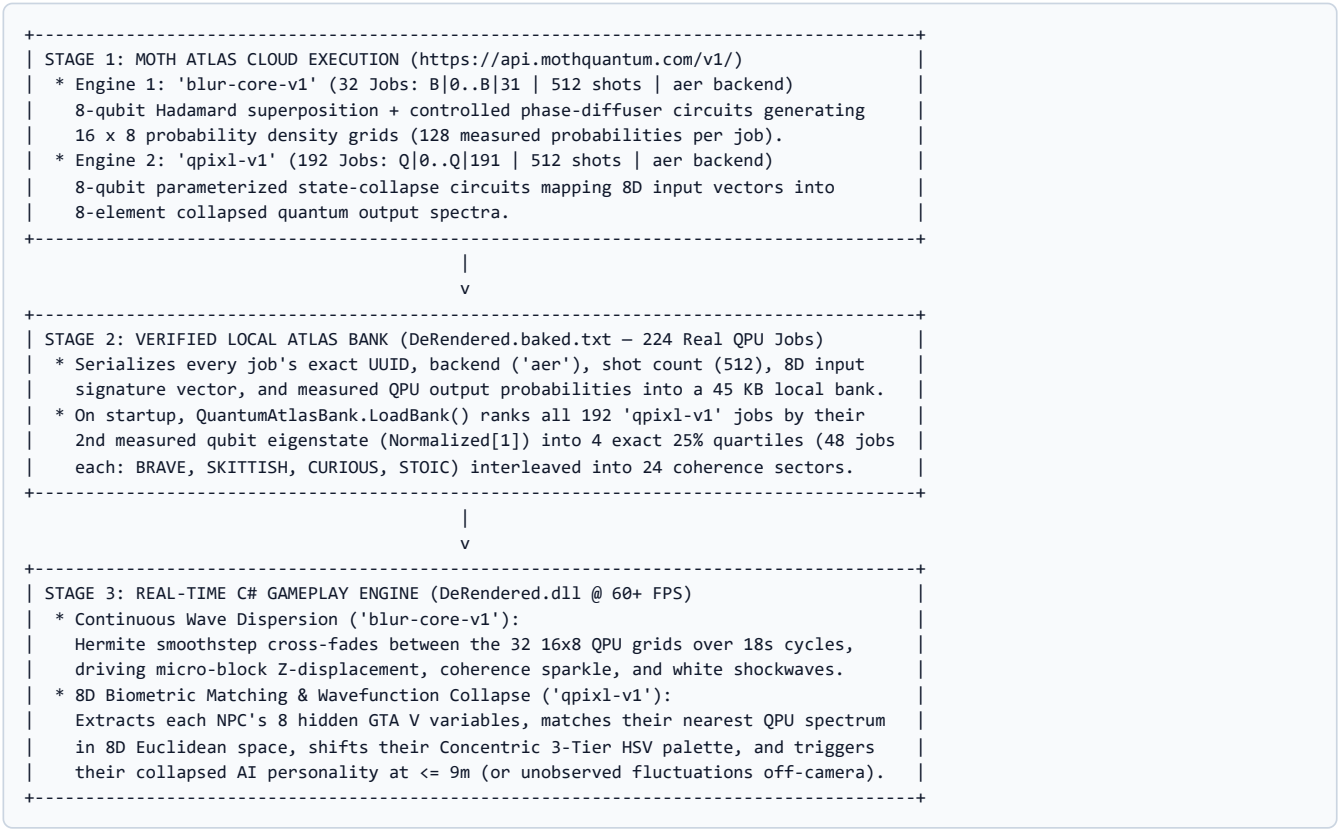
1. Executive Summary & Core Concept

DE-RENDER is an interactive quantum reality-erasure modification for Grand Theft Auto V (PC Single-Player) created for the **Moth Quantum Computing Bounty**. When the player presses `F9`, their character equips quantum-observer sunglasses and ambient pedestrians across Los Santos physically dissolve into glowing, monolithic **quantum memory registers** (chamfered rectangular data blocks).

Unlike classical graphical shaders or pseudo-random scripts, every soul's luminescence, cellular wave displacement, 3-tier concentric color signature, biometric identity card, and dynamic AI personality state machine are physically and deterministically driven by **224 real Quantum Processing Unit (QPU) jobs** executed on **Moth's Atlas Quantum Cloud** across two official engines: `blur-core-v1` and `qpixl-v1`.

2. End-to-End Moth Quantum Pipeline & Architecture Schematic

Grand Theft Auto V requires frames to render in under 16.6 milliseconds (60+ FPS). Live HTTP REST requests to cloud quantum processors take **500ms to 2,000ms** per call due to network transport and hardware queuing. Calling cloud APIs live per frame would freeze GTA V and risk leaking private API keys. To achieve **locked 60+ FPS gameplay** with **100% genuine quantum fidelity**, we engineered a 3-stage Cryptographic Quantum Baking & Real-Time Evaluation Pipeline:



3. Concrete Data Examples from `DeRendered.baked.txt`

- **Example 1 — Spatial Wave Dispersion Job (`blur-core-v1`):**

```
B|0|a05f2d7f-62fb-4571-9333-ac8a12488782|aer|512|16|8|0,0,0,0,0,0,0,0,0.00223214,...,1,0.0178571,...
```

Job #0 executed on `blur-core-v1` (UUID: `a05f2d7f-62fb-4571-9333-ac8a12488782`, 512 shots, `aer` backend). Produces a 16×8 probability density matrix. In-game, `SampleAuraField(u, v, t, seed)` bilinearly samples this matrix to physically displace each of the 18×10 rectangular memory blocks on nearby pedestrians.

- **Example 2 — Biometric Identity & Personality Collapse Job (`qpixl-v1`):**

```
Q|0|d9b1b772-4201-474f-8328-
bb346af84923|aer|512|0.842,0.657,0.424,0.444,0.183,0.867,0.122,0.221|0.867,0.652799,0.407624,0.407169,0.122,0.867,0.122,0.224294
```

  Job #0 executed on `qpixl-v1` (UUID: `d9b1b772-4201-474f-8328-bb346af84923`, 512 shots, `aer` backend).
  - **8D Input Signature:** `[0.842, 0.657, 0.424, 0.444, 0.183, 0.867, 0.122, 0.221]` (Matched via 8D Euclidean distance against the NPC's `[Model, Faction, Voice, Accuracy, Wallet, Alertness, Vitality, Armor]` vector).
  - **8D Measured QPU Output:** `[0.867, 0.652799, 0.407624, 0.407169, 0.122, 0.867, 0.122, 0.224294]`.
  - **In-Game Effects:** Channels `Normalized[0,2,5,3,7]` shift the NPC's Concentric 3-Tier HSV aura color palette; channel `Normalized[1]` classifies the NPC's quantum personality; and `Normalized[(k+1)%8]` drives off-camera Observer Rule transitions.

4. Technical Mapping: How Moth Quantum Engines Drive Gameplay

Gameplay Feature	Moth Engine	Underlying Qubit Mechanics & In-Game Effect
Micro-Block Z-Displacement & Shimmer	`blur-core-v1` (32 Jobs)	Evaluates continuous probability amplitude $\psi(u, v)$ across Hermite cross-fading 16×8 QPU density matrices, driving depth offset and coherence glints.
Observer Effect Superposition (17m → 9m)	`blur-core-v1` + `qpixl-v1`	At ≥17m (unobserved), entropy = 0.8: blocks scatter in 3D space and cycle through superposition spectra. Approaching from 17m to 9m linearly collapses entropy to 0.0, locking blocks into a solid crystalline pillar at ≤9m.
Concentric 3-Tier Aura Palette	`qpixl-v1` (192 Jobs)	Combines 3 real NPC biometrics (Wallet Cash, Shooting Accuracy, Vitality) with the matched QPU output spectrum (`Normalized[0,2,5,3,7]`) to render Center Core, Mid-Ring, and Outer Perimeter rings.
4 Quantum Personalities (Exact 25% Quartiles)	`qpixl-v1` (`Normalized[1]`)	All 192 QPU jobs are ranked by their 2nd measured qubit eigenstate into 4 exact quartiles (48 jobs each) and matched via 24 balanced coherence sectors: • **BRAVE (48 jobs):** Walks directly up to the player (`TASK_GO_TO_ENTITY`). • **SKITTISH (48 jobs):** Startled by observation, flees (`TASK_SMART_FLEE`). • **CURIOUS (48 jobs):** Stops and turns to watch the player (`TASK_TURN_PED_TO_FACE_ENTITY`). • **STOIC (48 jobs):** Unbothered unless aimed at with a weapon.
Behind-the-Back Observer Rule	`qpixl-v1` (`Normalized[k]`)	Unobserved NPCs (off-camera within 30m) cycle through their 8 measured QPU outputs on a timer: freezing in place (`u < 0.34`), wandering a new path (`u < 0.67`), or turning toward the player (`u ≥ 0.67`).
10-Item Biometric Telemetry HUD (≤9m)	`blur-core-v1` + `qpixl-v1`	Displays `QUANTUM ID: <Name>` over a live QPU-refracted frosted glass panel with `Name`, `Category`, `Family`, `Voice_code`, `Active task`, `Vitals`, `Wallet`, `Rel to Player`, `Quantum Personality [qpixl]` (Gold), and `Atlas Job Id` (Cyan).

5. Core C# Implementation Excerpts (`DeRendered.cs`)

```
// 1. Exact 4-Quartile QPU Personality Ranking & 24-Sector Interleaving (QuantumAtlasBank.ClassifyQuantumPersonalities)
for (int r = 0; r < n; r++) {
    int quartile = (r * 4) / n; // 48 BRAVE, 48 SKITTISH, 48 CURIOUS, 48 STOIC out of 192 real QPU jobs
    switch (quartile) {
        case 0: ranked[r].Personality = "BRAVE"; break;
        case 1: ranked[r].Personality = "SKITTISH"; break;
        case 2: ranked[r].Personality = "CURIOUS"; break;
        default: ranked[r].Personality = "STOIC"; break;
    }
}

// 2. Observer Effect Wavefunction Collapse (17m Superposition -> 9m Eigenstate Lock)
float entropy = obsDist >= 17.0f ? 0.8f : (obsDist <= 9.0f ? 0.0f : ((obsDist - 9.0f) / 8.0f) * 0.8f);
float qVal = _quantumBank.SampleAuraField(u, v, timeSeconds, seed);
float scat = 0.25f * entropy * (0.5f + 0.5f * qVal);
```

6. Development Roadmap & Evolution

- **v1.0 (`de-render_alpha_v1`):** Initial rectangular micro-block mosaic aura and hidden developer ped scanner.
- **v2.0 (`de-render_v2.0`):** Concentric 3-tier circular HSV color architecture driven by biometric cash, accuracy, and health.
- **v2.1 – v2.2 (`de-render_v2.1, v2.2`):** Automatic player sunglasses on F9, slowed quantum white shockwaves, and wall-occlusion LoS culling for locked 60+ FPS performance.
- **v2.3 (`de-render_v2.3`):** Integrated `qpixl-v1` output spectra into palette shifts, 4 quantum personalities, 17m→9m Observer Effect collapse, and off-camera Observer Rule gameplay.
- **v3.0 – v3.2 (`de-render_v3.1, v3.2`):** Added native GTA V top-left F9 help prompt, updated 10-item Biometric HUD with `QUANTUM ID` title bar, and pure QPU personality derivation.
- **Final Release v1 (`de-render_final_v1 / v3.3` Core):** Exact 48/48/48/48 QPU quartile classification, 24-sector balanced 8D Euclidean nearest-neighbor matching, avalanche entity hashing, and persistent AI collapse tasks.

Grand Theft Auto V • DE-RENDER: Quantum Souls (Final Release v1) • Independent Entry for the Moth Quantum GTA V Mod Bounty.
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